

Lesson 13: Zeros of Polynomial Functions

Assessment Criteria

- **Identification of Real Zeros** (20%): The student accurately identifies all real roots, utilizing techniques like the rational zero theorem and synthetic division.
- **Handling Complex/Irrational Zeros** (20%): The student uses the quadratic formula or factoring techniques to identify complex or irrational roots.
- **Multiplicity and Behavior** (10%): The student correctly identifies the multiplicity of each zero and its effect on the graph (crossing vs. touching the x -axis).
- **Graphing Accuracy** (10%): The graph accurately represents the calculated zeros, end behavior, and proper intercepts.
- **Process and Justification** (30%): The student demonstrates a clear, logical, and well-organized method for finding the zeros (e.g., showing synthetic division steps).
- **Mathematical Communication** (10%): The student uses proper notation and explains the relationship between the zeros, factors, and the graph's x -intercepts.